Allen Yuan

Contact Information

Address: Department of Mathematics, Northwestern University,
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Research

Algebraic topology, homotopy theory, and interactions with algebraic K-theory and arithmetic geometry.

Education & Employment

Assistant Professor 2024 - Present
Northwestern University, Evanston, IL

Member 2023 - 2024
Institute for Advanced Study, Princeton, NJ

NSF Postdoctoral Fellow 2020 - 2023
Columbia University, New York NY

Visitor 2020 - 2021
Institute for Advanced Study, Princeton, NJ

Ph.D. Mathematics 2015 - 2020
Massachusetts Institute of Technology, Cambridge, MA
Dissertation: *The Frobenius in higher algebra*, advised by Jacob Lurie.

A.B. in Mathematics, secondary in Computer Science 2011 - 2015
Harvard University, Cambridge, MA

Grants Fellowships & Awards

- NSF Topology Standard Grant 2025-2027
- NSF Mathematical Sciences Postdoctoral Research Fellowship 2020 - 2023
- NSF Graduate Research Fellowship 2015 - 2020
- International Mathematics Olympiad, Silver Medal 2010
- USA Mathematics Olympiad, 1st Place 2010

Papers

15. *Maps between spherical group rings* (with S. Carmeli and T. Nikolaus), 2024.
14. *Relative cyclotomic structures and equivariant complex cobordism* (with A. Blumberg and M. Mandell), 2023.
13. *G-spectra of cyclic defect*, (with T. Feng and D. Treumann), 2023.
12. *The sphere of semiadditive height* 1, **IMRN** (2023).
11. *The Chromatic Nullstellensatz* (with R. Burklund and T. Schlank), to appear in **Annals of Mathematics** 2022.
10. *Examples of chromatic redshift in algebraic K-theory*, **Journal of the European Mathematical Society**, (2024).
9. *Chromatic convergence for the algebraic K-theory of the sphere spectrum* (with A. Blumberg and M. Mandell), to appear in **Journal of the London Mathematical Society**, 2021.

8. *Higher semiadditive Grothendieck-Witt theory and the $K(1)$-local sphere* (with S. Carmeli), **Communications of the American Mathematical Society**, 3 (2023).
7. *A version of Waldhausen's chromatic convergence for TC* (with A. Blumberg and M. Mandell), **Bulletin of the London Mathematical Society**, 55 (2021).
6. *Wilson Spaces, Snaith Constructions, and Elliptic Orientations* (with H. Chatham and J. Hahn), **Inventiones Mathematicae**, 1 (2024).
5. *Integral Models for Spaces via the Higher Frobenius*, **Journal of the American Mathematical Society** 36 (2023).
4. *Exotic Multiplications on Periodic Complex Bordism* (with J. Hahn). **Journal of Topology** 13 (2020).
3. *Multiplicative Structure in the Stable Splitting of $\Omega SL_n(\mathbb{C})$*, (with J. Hahn), **Advances in Mathematics** 348 (2019).

Pre-graduate school:

4. *Irreducible Canonical Representations in Positive Characteristic* (with B. Gunby and A. Smith), **Research in Number Theory** 1(1) (2015).
3. *Proof of a Conjecture of Guy on Class Numbers* (with L. Chua, B. Gunby, and S. Park), **International Journal of Number Theory**, 11(4) (2015).
2. *Linearly Many Faults in Arrangement Graphs* (with E. Cheng and L. Liptak), **Networks**, **61(4)** (2013).
1. *Linearly Many Faults in (n, k)-Star Graphs* (with E. Cheng and L. Liptak), **Int. J. Found. Comp. Sci.** 22(7) (2011).

Invited Talks

2025: Brown University colloquium, Chicago chromatic homotopy theory weekend, MIT topology seminar

2024: Graduate Student Geometry and Topology Conference, Minnesota Topology Seminar, Chicago Algebraic Topology Seminar, Midwest Topology Seminar (IU), Princeton Topology seminar, MSRI Hot Topics workshop (Telescope conjecture)

2023: UCSD Colloquium, Duke Colloquium, UVA Colloquium, MIT Topology Seminar, Generalised Lie algebras in Derived Geometry (Utrecht, NL), Copenhagen Topology Seminar, Homotopy: Fruit of the fertile furrow (Cambridge, UK), Oberwolfach Homotopy Theory

2022: USC Colloquium, Northwestern Topology Seminar, Cornell Colloquium, AMS Western Sectional Meeting, UPenn G&T Seminar, ECHT, UCSD Topology Seminar, Chicagoland Topology Seminar, Utrecht Chromatic Homotopy Theory Seminar

2021: EAKTS, Jerusalem Topology Seminar, Munster Topology Seminar

2020: Johns Hopkins Topology Seminar, CUNY G&T Seminar, Oberseminar Regensburg: Integral Homotopy Theory (after Allen Yuan), EPFL Topology Seminar, Bonn Topology Seminar

2019: USC Topology Seminar, UCLA Topology Seminar, AMS Sectional Meeting, UVA Topology Seminar, Oberwolfach Seminar (TC and Arithmetic), Copenhagen Topology Seminar

2018: BIRS-CMO Meeting, Northwestern Topology Seminar, UChicago Topology Seminar, UIC Topology Seminar

Teaching

Northwestern University
Instructor for Math 220-2: Single variable integral calculus — Fall 2024
Instructor for Math 440-2: Introduction to Algebraic Topology — Winter 2025

Columbia University
Instructor for Math UN1102: Calculus II — Spring 2023
Instructor for Math UN2010: Linear Algebra — Spring 2022

Massachusetts Institute of Technology
Teaching Assistant for 18.03: Differential Equations — Spring 2018

MIT MathROOTS
Academic mentor — Summer 2019
Academic mentor — Summer 2018

MIT Directed Reading Program
Mentor for Alexander Clifton and Miguel Young — Winter 2015 - 2016

Harvard University
Course Assistant for Math 123: Rings and Fields — Spring 2014

IDEA Math
Instructor — 2011 - 2013

Professional Activities & Service

Northwestern Topology Seminar
Organizer — Fall 2024 - present

Columbia Algebraic Topology Seminar
Organizer — Spring 2022 - Spring 2023

MIT Juvitop
Organizer — Fall 2016

MIT Topology seminar
Organizer — Spring 2016

Referee for: Adv. Math., Isr. J. Math, JAMS, PAMS, Compositio, Selecta

Expert opinions for: IMRN, J. Top., Compositio, JAMS, Forum Pi, G&T, ASENS